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Practical Exercise 1

Part A

1. Total resistance

Circuit a

$$R = 20\ \Omega + 30\ \Omega + 50\ \Omega$$

$$= \underline{\underline{100\ \Omega}}$$

Circuit b

$$\frac{1}{R} = \frac{1}{20}\ \Omega + \frac{1}{100}\ \Omega + \frac{1}{50}\ \Omega = \frac{5 + 1 + 2}{100} = \frac{8}{100}$$

$$\frac{1}{R} = \frac{8}{100} \Rightarrow \frac{8}{8} R = \frac{100}{8}$$

$$R = \underline{\underline{12.5\ \Omega}}$$

2. Total Current:

Circuit a

$$I = \frac{V}{R} \Rightarrow \frac{125}{100} = \underline{\underline{1.25\ A}}$$

Circuit b

$$I = \frac{V}{R} \Rightarrow \frac{125}{12.5} = \underline{\underline{10\ A}}$$

$$= 0.001 \times 25.1 = \frac{V}{0.001}$$

3. Current through each resistor.

circuit a

$$\cdot I_{20} = I_{30} = I_{50} = \underline{\underline{1.25A}}$$

circuit b

$$\cdot I_{20} = \frac{125}{20} = \underline{\underline{6.25A}}$$

$$\cdot I_{100} = \frac{125}{100} = \underline{\underline{1.25A}}$$

$$\cdot I_{50} = \frac{125}{50} = \underline{\underline{2.5A}}$$

4. Voltage drop through each resistor.

Circuit a

$$V_{20} = IR = 1.25 \times 20 = \underline{\underline{25V}}$$

$$V_{30} = 1.25 \times 30 = \underline{\underline{37.5V}}$$

$$V_{50} = 1.25 \times 50 = \underline{\underline{62.5V}}$$

Circuit b

$$V_{20} = V_{100} = V_{50} = \underline{\underline{125V}}$$

5. Power dissipated through each resistor.

Circuit a

$$P = VI$$

$$P_{20} = 25 \times 1.25 = \underline{\underline{31.25 \text{ W}}}$$

$$P_{30} = 31.5 \times 1.25 = \underline{\underline{46.875 \text{ W}}}$$

$$P_{50} = 62.5 \times 1.25 = \underline{\underline{78.125 \text{ W}}}$$

Circuit b

$$P_{20} = 125 \times 6.25 = \underline{\underline{781.25 \text{ W}}}$$

$$P_{100} = 125 \times 1.25 = \underline{\underline{156.25 \text{ W}}}$$

$$P_{50} = 125 \times 2.5 = \underline{\underline{312.5 \text{ W}}}$$